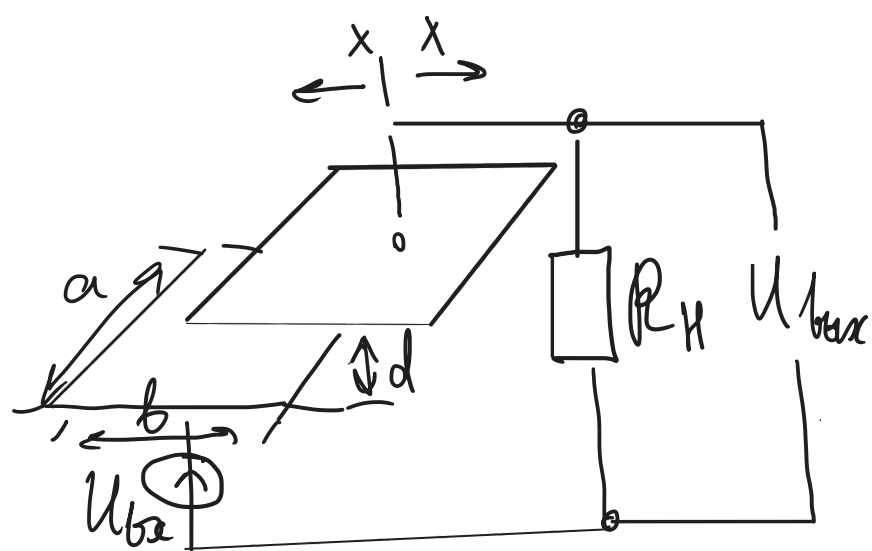




$$d = 0,01 \text{ mm}$$

$$a = 45 \text{ mm}$$

$$b = 45 \text{ mm}$$

$$U_0 = 36 \text{ B}$$

$$R_n = 400 \Omega$$

$$\omega = 10^5 \text{ rad/s}$$

$$\epsilon = 1$$

$$1) C = \frac{\epsilon \epsilon_0 S}{d} = \frac{\epsilon_0 a(b-x)}{d}$$

$$C(x) = \frac{8,85 \cdot 10^{-12} \cdot 0,045(0,045-x)}{0,01 \cdot 10^{-3}} = 39,625(0,045-x) \text{ nF}$$

$$\text{At } x = 0,03 \text{ m} = 30 \text{ mm} \quad C = 0,597 \text{ nF}$$

$$2) X_c = \frac{1}{\omega C} = \frac{d}{\omega \epsilon_0 a(b-x)}; Z(x) = \sqrt{R_n^2 + X_c^2} = \sqrt{R_n^2 + \frac{d^2}{(\omega \epsilon_0 a(b-x))^2}}$$

$$Z(0,03) = \sqrt{R_n^2 + \left(\frac{1}{\omega C(0,03)}\right)^2} = \sqrt{1600 + \left(\frac{1}{0,597 \cdot 10^{-4}}\right)^2} =$$

$$\approx \frac{1}{0,597 \cdot 10^{-4}} = 16750 \Omega$$

$$3) I(x) = \frac{U_0}{Z(x)}; I(0,03) = \frac{36}{16750} = 2,15 \text{ mA}$$

$$4) U_{0n}(x) = I(x) \cdot R_n$$

$$U_{0n}(0,03) = I(0,03) \cdot R_n = 2,15 \cdot 40 = 86 \text{ mV}$$

$X, \text{ mm}$	0	5	10	20	30	40	45
$C(x), \text{ nF}$	1,79	1,55	1,35	0,99	0,55	0,19	0
$Z(x), \Omega$	5,6	6,3	7,2	10,1	16,8	52,6	—
$I(x), \text{ mA}$	6,42	5,71	5	3,56	2,15	0,68	—
$U_{0n}(x), \text{ mV}$	256	228	200	142	86	27	—

